

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 03 — BUILDING YOUR FIRST PLC PROGRAM

Student Edition

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Lab 03: Building Your First PLC Program

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 4–5 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Create a new OpenPLC Ladder Diagram project.
- Define Boolean and Real variables for an industrial tank-fill process.
- Build logic to automatically fill a 0–10 ft tank between operating setpoints.
- Implement high/low alarm logic and an emergency-stop condition.
- Document and validate the intended control behavior before connecting to an HMI.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	plc	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

Variable Plan

Variable	Data Type	Purpose
TankFillAuto_Cmd	BOOL	Internal automatic fill latch/command
TankValveOpen	BOOL	Discrete command/status for the fill valve
Estop_Cmd	BOOL	Emergency-stop input; TRUE forces automatic fill to stop
TankLevelHi_Alarm	BOOL	High-level alarm state
TankLevelLo_Alarm	BOOL	Low-level alarm state
TankLevel_LIT	REAL	Measured tank level in feet
TankEmpty_SP	REAL	Automatic fill start setpoint (2.0 ft)
TankFull_SP	REAL	Automatic fill stop setpoint (8.0 ft)
TankLoAlarm_SP	REAL	Low-level alarm setpoint (1.0 ft)
TankHiAlarm_SP	REAL	High-level alarm setpoint (9.0 ft)

Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Start the PLC VM from a clean snapshot. Launch OpenPLC Editor and create a new project folder named TankFill.

Step 2. Create a new POU named TankFill and select LD (Ladder Diagram) as the language.

Step 3. Add the variables listed in the Variable Plan table in this manual. Use BOOL for discrete states/commands and REAL for tank-level values and setpoints.

Step 4. Set the operating values: TankEmpty_SP = 2.0 ft, TankFull_SP = 8.0 ft, TankLoAlarm_SP = 1.0 ft and TankHiAlarm_SP = 9.0 ft.

Step 5. Insert left and right power rails with enough pins for the complete program, then create the first logic rung.

Step 6. Build the valve-control rung: TankFillAuto_Cmd AND NOT TankLevelHi_Alarm energizes TankValveOpen.

Step 7. Build the automatic-start rung: when TankLevel_LIT <= TankEmpty_SP, SET TankFillAuto_Cmd.

Step 8. Build the automatic-stop/safety rung: when TankLevel_LIT >= TankFull_SP OR Estop_Cmd is TRUE OR TankLevelHi_Alarm is TRUE, RESET TankFillAuto_Cmd.

Step 9. Build the high-level alarm rung: TankLevel_LIT >= TankHiAlarm_SP energizes TankLevelHi_Alarm.

Step 10. Build the low-level alarm rung: TankLevel_LIT <= TankLoAlarm_SP energizes TankLevelLo_Alarm.

Step 11. Check every variable name and data type. Confirm that all contacts/coils/expressions are wired to complete rungs and that no instruction is floating.

Step 12. Save the project. If the editor provides validation/compile checking in your VM image, run it and correct any errors before submission.

Step 13. Create a logic test table covering at least six scenarios: empty tank, normal filling, full tank, high alarm, low alarm and emergency stop.

Step 14. Capture screenshots of the variable table and the completed ladder logic. Submit the project folder together with your written report.

Project Files and required snapshots are available in screenshots folder

Logic Test Table (Step 13)

Scenario	TankEmpty	TankFull	TankHighAlarm	TankLowAlarm	Estop_Cmd	TankFillAuto_Cmd	TankValveOpen	Hi Alarm	Lo Alarm
Empty Tank	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
Normal Filling	FALSE	FALSE	FALSE	FALSE	FALSE	TRUE	TRUE	FALSE	FALSE
Full Tank	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE
High Alarm	FALSE	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE
Low Alarm	TRUE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	FALSE	TRUE
Emergency Stop	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE			

Short Explanation for Report

The TankFill PLC program automatically controls tank filling. When the tank becomes empty (TankEmpty), the system sets TankFillAuto_Cmd, opening the inlet valve through TankValveOpen. Filling continues until a stop condition occurs. If the tank reaches full level (TankFull), a high-level alarm activates (TankLevelHi_Alarm), or an emergency stop (Estop_Cmd) is pressed, the automatic fill command is reset and the valve closes. Separate alarm rungs generate high-level and low-level alarm indications for operator monitoring.

Knowledge Check

1. Why are BOOL and REAL appropriate data types for this project?

Response: BOOL data types are used for discrete conditions that have only two states (TRUE/FALSE), such as TankFillAuto_Cmd, Estop_Cmd, and alarm signals. REAL data types are used for continuous numeric values such as tank level measurements and setpoints (TankLevel_LIT, TankEmpty_SP, TankFull_SP) because they can store decimal numbers.

2. What is the purpose of a SET/RESET pair for TankFillAuto_Cmd?

Response: The SET/RESET pair creates a latch. The SET instruction turns TankFillAuto_Cmd ON when the tank becomes empty, and the RESET instruction turns it OFF when a stop condition occurs, such as a full tank, high-level alarm, or emergency stop.

3. What should happen if the tank reaches 9 ft while the fill valve remains commanded open?

Response: At 9 ft, the high-level alarm condition is reached. TankLevelHi_Alarm should become TRUE, causing TankFillAuto_Cmd to reset and the fill valve (TankValveOpen) to close immediately to prevent tank overfilling.

4. Why should PLC logic be validated in simulation before use with physical equipment?

Response: Validating PLC logic in simulation helps identify programming errors, unsafe operating conditions, and incorrect control actions before deployment. This reduces the risk of equipment damage, process disruptions, and safety incidents while ensuring the control strategy operates as intended.